DWPose Repository Workflow

Effective Whole-body Pose Estimation with Two-stages Distillation — training & deployment overview

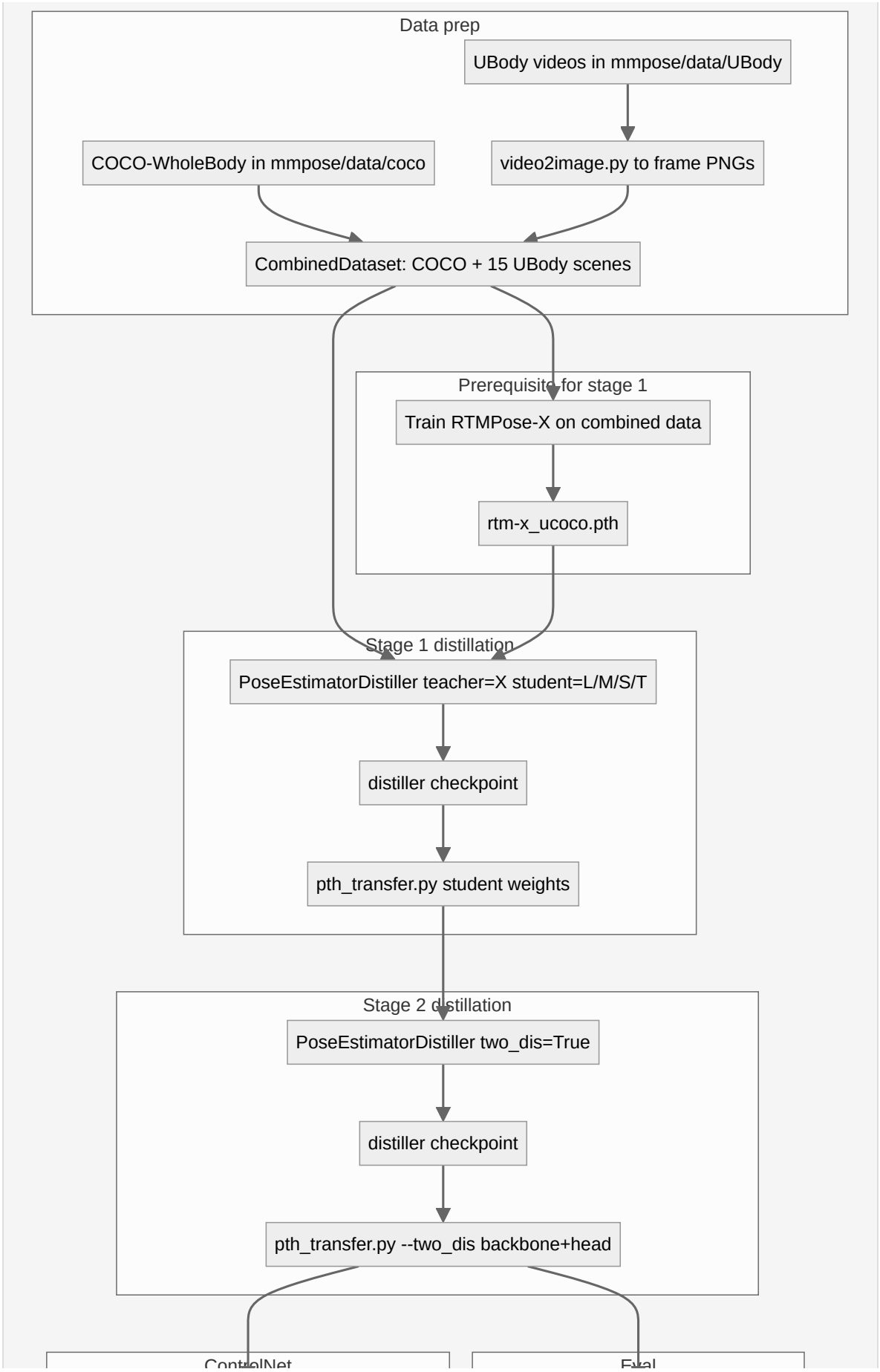
Repo at a glance

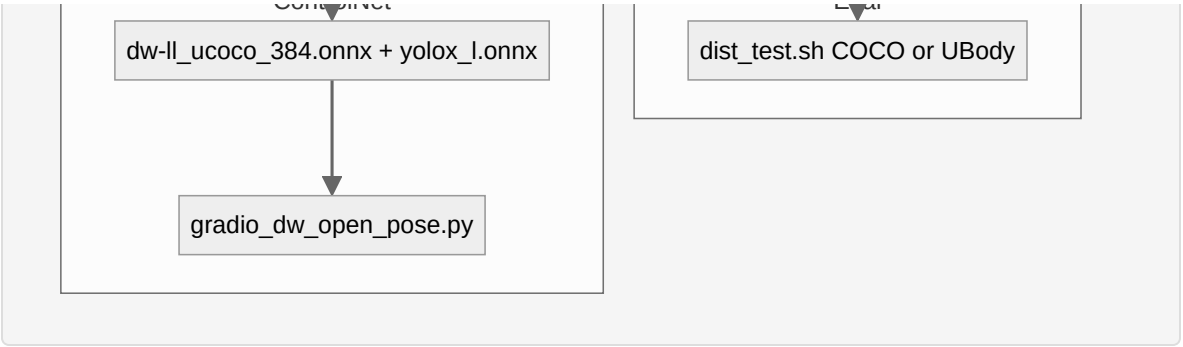
DWPose splits into two largely independent stacks:

Area	Path	Role
Training & eval	`mmpose/`	MMPose fork + custom two-stage distillation (`PoseEstimatorDistiller`)
ControlNet / demos	`ControlNet-v1-1-nightly/`	ONNX YOLOX + DWPose → OpenPose-style maps for ControlNet

Goal: strong whole-body pose (133 keypoints) on **COCO-WholeBody + UBody**, compressed via teacher→student distillation in two stages.

End-to-end flow


Data prep
UBody videos in mmpose/data/UBody
COCO-WholeBody in mmpose/data/coco
video2image.py to frame PNGs
CombinedDataset: COCO + 15 UBody scenes
Prerequisite for stage 1
Train RTMPose-X on combined data
rtm-x_ucoco.pth
Stage 1 distillation
PoseEstimatorDistiller teacher=X student=L/M/S/T
distiller checkpoint
pth_transfer.py student weights
Stage 2 distillation
PoseEstimatorDistiller two_dis=True
distiller checkpoint
pth_transfer.py --two_dis backbone+head
ControlNet
Eval



1. Data preparation

- Place **COCO** under `mmpose/data/coco`.
- Place **UBody** under `mmpose/data/UBody`.
- Run `cd mmpose && python video2image.py` (ffmpeg, 30 fps) to convert UBody videos to images.
- For UBody evaluation: `python add_cat.py` on annotations.

Training configs build a `CombinedDataset`: COCO-WholeBody train plus 15 UBody scenes, with top-down augmentation and SimCC targets. Default schedule: **270 epochs**, EMA, cosine LR, pipeline switch in the last 30 epochs.

2. Teacher model (stage 1 prerequisite)

Stage-1 distillation expects a frozen **RTMPose-X** checkpoint, e.g. `work_dirs/rtmpose-x_8xb64-270e_coco-ubody-wholebody-256x192/rtm-x_ucoco.pth`.

```
cd mmpose
bash tools/dist_train.sh configs/wholebody_2d_keypoint/rtmpose/ubody/rtmpose-x_8xb64-270e_coco-ubody-wholebody-256x192.py 8
```

`tools/dist_train.sh` launches distributed `tools/train.py` (MMEngine Runner).

3. Stage 1 distillation

```
cd mmpose
bash tools/dist_train.sh
configs/distiller/ubody/s1_dis/rtmpose_x_dis_l__coco-ubody-256x192.py 8
```

Config: `PoseEstimatorDistiller` — teacher (large, frozen) + student (smaller); dataloaders from student whole-body UBody config.

Loss	Role

`loss_kpt`	SimCC / KLDiscretLoss on student features
`loss_fea`	MSE on last backbone features (1×1 align if channels differ)
`loss_logit`	KL on SimCC x/y logits (teacher vs student)

Stage 1 scales distillation terms by `(1 - epoch/max_epochs)`.

```
python pth_transfer.py $dis_ckpt $new_pose_ckpt
```

Exports `student.*` weights only (standard TopdownPoseEstimator checkpoint).

4. Stage 2 distillation

```
cd mmpose
bash tools/dist_train.sh configs/distiller/ubody/s2_dis/dwpose_l-ll__coco-
ubody-256x192.py 8
```

- `two_dis=True` — teacher is stage-1 distilled model; same architecture as student.
- Shorter run (e.g. 60 epochs); feature loss off; stronger logit KD.
- Supervision uses `student.head` on **teacher backbone** features — head refinement.

```
python pth_transfer.py $dis_ckpt $new_pose_ckpt --two_dis
```

Merges `teacher.backbone` + `student.head` for inference.

```
bash tools/dist_test.sh configs/wholebody_2d_keypoint/rtmpose/ubody/rtmpose-
l_8xb64-270e_coco-ubody-wholebody-256x192.py $pose_ckpt 8
```

5. ControlNet inference (ONNX)

- Detector: `yolox_l.onnx`
- Pose: e.g. `dw-ll_ucoco_384.onnx` in `ControlNet-v1-1-nightly/annotator/ckpts/`
- `annotator/dwpose/wholebody.py` — ONNXRuntime, OpenPose keypoint layout

```
cd ControlNet-v1-1-nightly
python gradio_dw_open_pose.py
python dwpose_infer_example.py
```

Summary

1. Supervised whole-body training on COCO + UBody (RTMPose / SimCC).
2. Stage 1: large teacher → small student (feature + logit KD + GT on full student).
3. Stage 2: fixed distilled backbone; tune head with logit KD + GT.
4. `pth_transfer.py` → deployable MMPose weights.
5. ControlNet uses exported ONNX models.